Comparative Study of Machine Learning Regression Models for Student Performance Prediction

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Abstract — Since the 2000's a huge number of studies have emerged in the area of educational data mining that aims to help students by predicting the academic performance to flag students who may need help at school and university. This paper aims to compare and evaluate the performances of four supervised machine learning regression models, namely Linear Regression, Ridge Regression, Decision Tree Regressor and Random Forest Regressor, to find the best predictive model for student's grades. The UCI Student Performance Dataset (student-mat.csv) having a total of 395 student records further with 33 attributes representing demographic, social, and academic details with final mathematics grade G3 as target variable is used. The categorical variables were processed using LabelEncoder, data was divided into 80:20 with train-test-split and StandardScaler was used to normalize the data. The evaluation of all four models used MAE, MSE and R^2 score. The best model we found is the Random Forest Regressor with an R^2 of 0.8288 a MAE of 1.147 and a MSE of 2.838. Thus, we can conclude that this model can explain 83% of the variation in student final grades. Feature importance analysis revealed that G1 and G2, which are grades obtained in the first and second periods respectively, are the strongest predictors of G3. This validates the importance of interim records for building effective early warning systems.

Keywords — machine learning, student performance prediction, regression models, random forest, educational data mining, UCI dataset, feature importanc**e**

I. INTRODUCTION

The adoption of technology in education has changed how schools and universities collect, store and analyse students' data. In the past, institutions collected evidence/ information in paper format or networked systems through direct observation. Now they have at their disposal, at a click of a mouse, a massive amount of data pertaining to almost every aspect of a student's academic life- Background, family situation, attendance, social behaviour, study habits, performance, etc. This shift opens up powerful possibilities for how teachers understand and respond to their students.

Educational data mining (EDM) is the use of computational methods in order to build models or discover patterns from data that originate in an educational context. In the past two decades, the field has become rich and diverse. Computer science, statistics, psychology, and education have all played a part. Today, serious interest is growing globally.

One of the most practically relevant challenges in EDM is predicting student academic achievement. If historical data can tell us how a student will perform before the course ends, it allows for valuable early intervention. An educator who can identify at the beginning of the semester that a student will struggle can offer additional support. A meeting can be arranged by a counsellor on early warning. A school can provide scholarships, tutoring, or mental health support before a student reaches a crisis point. For years, schools have monitored performance via periodic tests and final examinations. However, by the time a low performance appears on the grades, weeks or months of intervention time has already passed. Machine learning allows us to switch from looking back to looking forward.

Machine learning algorithms learn patterns in data and use those patterns to make predictions about new instances. Unlike rule-based programming that needs to understand mechanisms, the models detect patterns from examples. This allows them to detect complex combinations of behavioural, social and academic signals. When our outcome can take infinitely many values (e.g. a final numeric grade), the choice of model is a regression-based model. Such models will provide exact predicted grades (not just pass/fail labels) and will hence be more useful.

This paper discusses the four most famous regression algorithms which include: Linear Regression, Ridge Regression, Decision Tree Regressor and Random Forest Regressor. They will be put to the test using the same dataset under the same conditions to offer a direct comparison. We have utilized the UCI Student Performance Dataset [9], collected by Cortez and Silva [1], which contains information about 395 secondary school students from Portugal with 33 features and the final mathematics grade G3 as the target variable.

Through this work, we present (1) reproducible and fair comparison of four regression models, (2) evaluation with MAE, MSE and R^2, (3) importance of features analysis and (4) insights useful for the building of machine-learning-based student support systems. The literature review is covered in section II, section III describes the dataset, section IV describes the methodology, section V describes the models, section VI states the results, section VII discusses the findings and section VIII concludes and states the way forward.

II. LITERATURE REVIEW

In the last 20 years, one can find widespread application of data mining and machine learning techniques in academia. Cortez and Silva [1] conducted the first and most cited study that applied and compared several machine learning algorithms. These include decision trees, random forests, neural networks, and support vector machines to predict student marks in the same student-mat.csv dataset used in this paper. The findings demonstrated that the integration of G1 and G2 as intermediate period grades led to notable enhancements in prediction accuracy across all models. If these features were excluded from the model, accuracy dropped significantly, suggesting that previous academic performance is an impactful predictor. This work has set the dataset as a benchmark in the field.

Yadav, Bharadwaj and Pal [2] applied decision tree algorithms to predict whether students will be retained or drop out based on data from an Indian university. Better past exam scores and attendance rates are much more informative than demographic features such as gender and family income. Also, the study suggested that decision trees are easy to visualize and interpret, thus making the outputs understandable and trusted by teachers and administrators without ML backgrounds. Decision trees become unstable on small datasets, thus motivating research into ensemble methods.

A comparative study on student behaviours using available data from a Jordanian university was carried out by Amrieh, Hamtini, and Aljarah [3]. They compared Naive Bayes, Neural Networks, and Decision Trees along with an ensemble technique like Random Forest and AdaBoost. According to their results, ensemble methods beat single models on all metrics. Interestingly, the meaningful predictor of academic performance was found to be behavioural signals, including raising hands and participating in class, which can be used as additional features along with grades.

Delen [4] undertook a large-scale analysis of records involving more than 26,000 students at a large American university to predict student retention. The ensemble decision tree methods achieved the best results among various machine learning techniques. Further, the study showed that in actual institutional deployments, model interpretability matters just as much as accuracy – a model that is just slightly less accurate but easier to explain will be more useful in practice because it earns educators' trust who act on its predictions.

The research conducted by Marquez-Vera, Romero, and Ventura [10] was centred on school failure and dropout prediction. Careful selection of the features had a bigger impact on performance than the choice of algorithm during modelling. Educational datasets often contain many redundant and weakly predictive features. Remove these features prior to training for substantial improvements in accuracy and efficiency. This finding played a direct role in the present research' feature importance analysis.

In a broad survey of educational data mining, Romero and Ventura [11] reviewed hundreds of studies on the subject. In their opinion, the student performance prediction problem is the most studied one. To make the studies comparable they demanded standard benchmarking practices. As the majority of studies that have been published were on classification than regression, this implies that prediction of continuous grades as by regression has not been conducted as widely as it could have been.

According to the aggregate work carried out, machine learning is an efficient tool for predicting the academic performance of students. When it comes to best results, techniques like Random Forest always work best. Nevertheless, most existing comparative studies differ in their preprocessing pipelines or use proprietary datasets, prohibiting direct comparison.

In this paper, we tackle these shortcomings through the use of a fully standardized experimental pipeline over a publicly available dataset.

III. DATASET DESCRIPTION

A. Source and Overview

The UCI Student Performance Dataset [9] used in this study was collected by Cortez and Silva [1] as part of a study investigating factors affecting student performance in Portuguese secondary schools. Data was gathered during the 2005–2006 academic year from two schools — Gabriel Pereira and Mousinho da Silveira — through school administrative records and student questionnaires. The mathematics course subset (student-mat.csv) contains 395 records with 33 attributes.

B. Feature Categories

There are three sets that the features fall into. The first includes the basic details like age, sex, type of the address, family size and parental cohabitation. The student's basic personal details are described in this group.

The second and most extensive set of attributes includes socioeconomic and behavioural variables. These include parent education level (Medu, Fedu) parent occupation (Mjob, Fjob), weekly study time, number of past course failures, club memberships, internet access, relationship status, family relationship quality, free time, social outings, and alcohol consumption on weekdays and weekends (Dalc, Walc). This captures the resources that shape the student's learning environment.

The third group consists of academic attributes, these are school and family educational support, the number of absences throughout the year and especially the first and second period grades G1 and G2. These two grades are of particular interest as they constitute the most immediate signal of a student's academic trajectory and as confirmed by this study, they are the strongest predictors of the final grade G3.

C. Target Variable

The target variable G3 is the final period mathematics grade, ranging from 0 to 20 as a continuous integer. Grades are concentrated in the 10–15 range. A notable group of students received a grade of 0, which in the Portuguese system typically indicates course withdrawal rather than academic failure. This distinction makes regression modelling of G3 somewhat complex, as zero grades differ qualitatively from low but non-zero grades.

D. Data Quality Assessment

A quality check was performed before any model training using df.isnull().sum() on all 33 columns. All 395 student records were found complete with no missing values across all 33 attributes. This means no data imputation was performed, and any differences observed in model performance can be attributed purely to algorithmic behaviour rather than any data filling. The cleanliness of this dataset is one of the reasons it is so widely used as a benchmark in educational data mining research.

IV. METHODOLOGY

To ensure no advantage or disadvantage of any model, the experimental pipeline was designed such that each stage could be performed similarly on all four models. Experiments were conducted in Google Colab using Python 3. The primary libraries were: Pandas to manipulate data, NumPy to perform numerical computations, Scikit-learn to implement model and Matplotlib to visualize [8].

A. Data Loading and Exploration

The student-mat.csv file makes use of semicolons as separators. It was loaded using pd.read_csv() with sep=';'. The standard functions exploration – df.head(), df.shape, df.columns and df.info() – confirmed 395 rows and 33 columns, identified 16 categorical features, and confirmed no missing values. This step helped us identify if any errors were encountered while loading the dataset and what preprocessing would be required.

B. Categorical Encoding

Machine learning models require numeric inputs, so all 16 categorical columns like sex, address type, parental occupations, and binary yes/no responses were encoded using Scikit-learn's LabelEncoder. This assigns each unique category an integer value, for example "rural" = 0 and "urban" = 1. While LabelEncoder imposes an implied order that may not always reflect true relationships, it is the standard accepted approach for exploratory comparative studies of this kind.

C. Feature and Target Definition

After encoding, we created the feature matrix X by deleting the G3 column using df.drop('G3', axis=1), and the target vector y as df['G3']. X retained both G1 and G2 in order to reflect a realistic deployment situation where a school has mid-year results and wants to know what the final performance would be before the end of term. This setup matches how an early warning system would behave in reality.

D. Train-Test Split

The train_test_split() function was used to split the data into 80% training and 20% test with a random seed of 42. Thus we got 316 training samples and 79 testing samples. The random state was set for complete reproducibility. The accepted norm in the literature is to use 80:20 for training and testing dataset which takes care of having enough data for training and quality test evaluation.

E. Feature Scaling

Feature scaling makes sure that all the features equally impact the learning process of the model. If we don't apply feature scaling, those features that have a broad range of values, like absences (from 0 to 93), will take the lead in machine learning algorithms that consider the value scale of the features. For this purpose we utilized StandardScaler, which brings the values of the features closer to zero and sets their standard deviation to 1. Importantly, fitting was done only on the training dataset with fit_transform().

V. MACHINE LEARNING MODELS

A. Linear Regression

Linear Regression is an algorithm that builds a model to represent the relationship between inputs and the output (dependent variable) by using a linear equation which can be stated as $y = w_1x_1 + w_2x_2 + ... + w_nx_n + b$. The weight values 'w' are coefficients while 'b' stands for the intercept value. It computes the optimal weight parameters through the ordinary least squares (OLS) method, which calculates the minimum difference between the expected and real grades without any iteration process.

The advantages of linear regression include its speed and simplicity. The linear equation allows us to see which features received the highest weights and, therefore, the biggest impact on predictions. This can be useful in an education context, where teachers and administrators want to know why a student received a given prediction. The assumption of linearity is its main limitation which might not hold in various educational datasets as many features interact with one another in complex non-linear ways.

B. Ridge Regression

Ridge Regression was created to solve the issue faced by linear regression where having correlated features leads OLS to give very large and unstable coefficient values. Ridge mitigation methods apply to the cost function a penalty (L2 norm) proportional to the sum of the squares of the weights times a factor called lambda. As a result, it prevents the algorithm from assigning large values to any single feature, leading to more stable predictions [6].

The main idea of Ridge Regression is that the feature weights will be well spread across various features and not rely too heavily on any one which can be noisy training data. The student performance dataset has G1, G2, study time, and fails that are correlated with each other. Thus, Ridge is a natural choice. Although, like Linear Regression, Ridge is still a linear model. Subsequently, it cannot represent non-linear relationships between features and target.

C. Decision Tree Regressor

Instead of using two linear models, Decision Tree Regressor uses a different approach. Instead of using one big equation on the entire dataset, it learns a series of local if- then rules that break the feature space into smaller regions. The algorithm performs a test on a feature at each internal node, for example, G2 > 12, dividing the data into two groups based on the answer. It assesses all potential features and split points in order to determine the one with the least prediction error. This process continues repeating until each leaf is a sufficiently small group [7].

Decision trees are fully non-parametric and capable of capturing highly intelligent non-linear relationships and generating highly human-readable models that can be read like a flowchart. Yet their most significant weakness is a tendency to overfit. In particular, small datasets where a fully grown tree memorises the training data rather than learns from it. Techniques like pruning or imposing maximum tree depth can reduce overfitting but it remains a fundamental drawback of single decision trees.

D. Random Forest Regressor

The Random Forest Regressor is an ensemble technique that was explicitly designed to deal with the issues of overfitting single decision trees. Random Forest fits hundreds of trees instead of training a single tree on the entire training set. Each tree gets a different random bootstrap sample of the training data, which is referred to as bagging. This guarantees that each tree learns different things. The average of all individual trees' predictions is the final prediction for the student [5].

Random Forest also randomly selects a subset of features to be utilized in each split of each tree, which ensures structural differences in trees along with differences in the data samples on which they were trained. This helps to lower the correlation between trees allowing the ensemble to be more robust. The combined effect greatly enhances generalization, and the variance reduces significantly without increasing bias. Thus, it generalizes far better than any individual tree. Another advantage, Random Forests have an in-built mechanism to evaluate the importance of variable features by monitoring the decrease in error of all trees.

VI. RESULTS AND EVALUATION

A. Quantitative Performance Comparison

After all the prior steps had been accomplished, all four regression models were trained on the training data set with 316 samples and evaluated on the test data set with 79 samples using the following three measures in order to measure their performance: Mean Absolute Error (MAE) is the average value of the absolute difference between predicted and actual grade values. Lower MAE value implies that the predicted values are closer to the real grade values on average. The Mean Squared Error (MSE) is the average value of the squared difference between predicted and actual values where larger errors are penalized more strongly compared to smaller errors. R^2 Score, otherwise known as the coefficient of determination, refers to the percentage of variance in G3 that is accounted for by the model: R^2=1.0 implies perfect fit, whereas R^2=0 implies that the model predicts the grade no differently than the mean grade for all individuals. The results for these models under investigation are listed below in Table I.

TABLE I. Performance Comparison of Regression Models

Model	MAE	MSE	R^2 Score	Rank
Linear Regression	1.421	3.872	0.7614	3rd
Ridge Regression	1.418	3.855	0.7625	2nd
Decision Tree	1.734	5.190	0.6817	4th
Random Forest	1.147	2.838	0.8288	1st

Among these, Random Forest Regressor produced the best results with an R^2 = 0.8288, MAE = 1.147 and MSE = 2.838. This means that it explained nearly 83% variance of the students' final grades. The actual grades were on average about 1.15 grade points away from the predicted grade which was on a 0-20 scale which is strong and practically useful. Ridge Regression (R^2 = 0.7625) is close enough to regular Linear Regression (R^2 = 0.7614). From these estimates, it can be stated that the level of multicollinearity among the independent variables is relatively low to cause instability in the OLS estimator. The Decision Tree Regressor model provides the worst estimate, where R^2 = 0.6817 and MSE = 5.190, consistent with its known tendency to overfit on small datasets.

B. Model Comparison Visualizations

Bar charts comparing all four models on R^2, MAE, and MSE are shown in Figures 1, 2, and 3 respectively. The R^2 chart immediately shows Random Forest as the clear winner, with Ridge and Linear Regression close together and Decision Tree noticeably lower. The MAE chart shows the reverse pattern (lower is better) with Random Forest having the shortest bar. In the MSE chart, the poor performance of the Decision Tree becomes even more obvious as squaring the errors amplifies its large individual mistakes, making its bar stand far above the others.

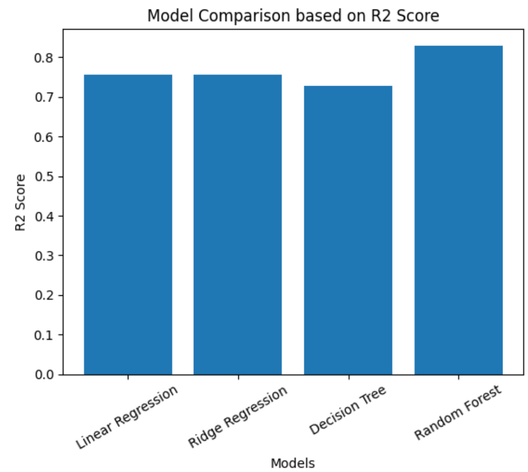

Fig. 1. Comparison of regression models based on R^2 Score

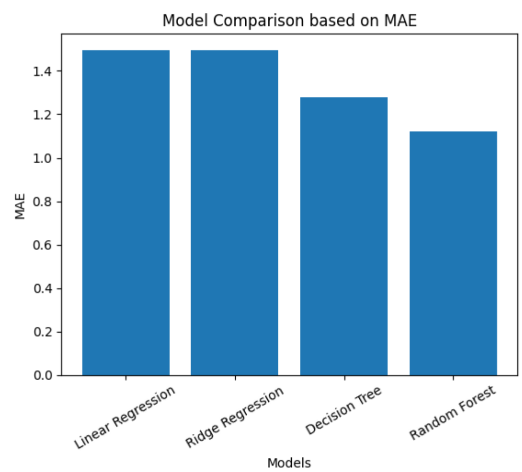

Fig. 2. Comparison of regression models based on MAE

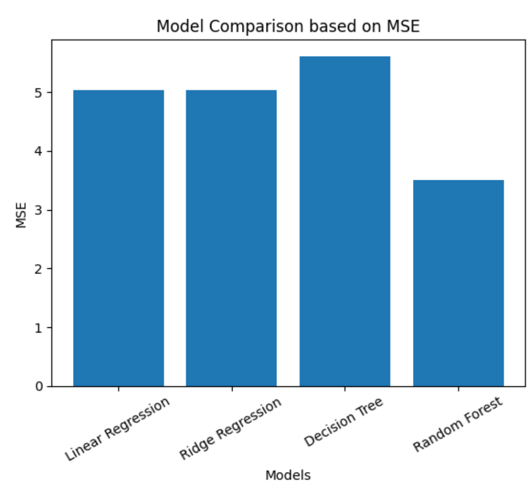

Fig. 3. Comparison of regression models based on MSE

C. Actual vs. Predicted Values

Figure 4 shows a scatter plot of actual G3 scores against predicted G3 scores for the Random Forest model on the 79 test students. Each point represents one student. The diagonal line is the line of exact predictions. It can be seen from the scatterplot that most dots cluster around the diagonal line, especially for those whose actual scores fall between 9-16 points, where the greatest density of dots is observed. In this mid-point score range, the predicted scores are usually very close, within one or two points of the real scores.

Prediction accuracy decreases towards either end of the scale, especially for those with G3 score being 0, because such low score indicates that the student dropped out of the course, thus the grade point is rarely encountered during learning.

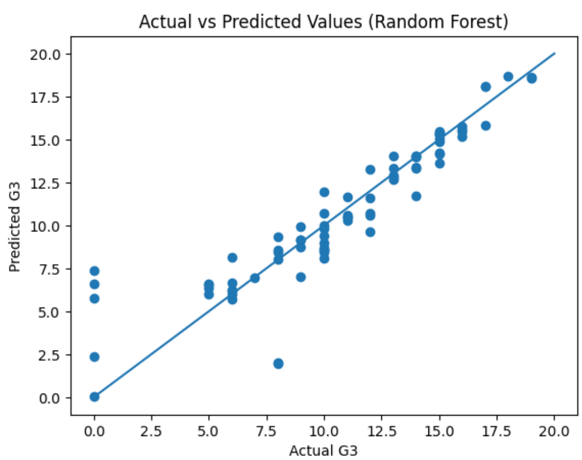

Fig. 4. Actual vs. Predicted G3 values — Random Forest Regressor

D. Feature Importance Analysis

One of the valuable properties of the Random Forest algorithm is its ability to measure the relative importance of each input feature, determined by how much each feature contributes to reducing prediction error across all trees in the forest. Features used more frequently to produce greater error reduction are considered more important. Figure 5 shows the top 10 most important features identified by the model.

The second period grade G2 is by far the most important feature with an importance score of approximately 0.80, meaning it alone drives the vast majority of the model's predictions. Absences come in a distant second at around 0.12. Age and reason for choosing the school follow with very small scores. G1 ranks fifth rather than second because G2 already captures most of the academic information G1 would provide, making G1 far less useful once G2 is included. Social and demographic features such as family relationship quality, mother's job, romantic status, health, and extracurricular activities appear in the lower ranks, confirming that direct academic signals are far more predictive than background factors.

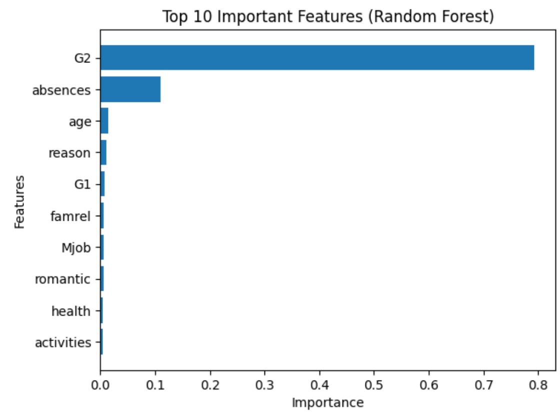

Fig. 5. Top 10 Important Features — Random Forest Regressor

VII. DISCUSSION

The experimental results present clear and informative observations about both the technical performance of the models and their practical applicability in educational settings. We discuss these observations first with respect to the algorithms themselves and then with respect to broader implications for educational institutions.

In terms of performance among the four models, Random Forest Regressor stands out significantly over the other models. With an R^2 score of 0.8288, it outperforms the second-best model, which is Ridge Regression with 0.7625, by more than 8%. On the educational side, this means more at-risk students will be accurately predicted, as well as higher precision in predicting grades for teachers and counsellors. This superiority of the algorithm stems from its ensemble approach; it trains hundreds of decision trees and averages their predictions. As a result, this technique allows us to reduce both the variance and overfitting compared to the single tree model, without sacrificing flexibility.

The near identical results of Linear Regression and Ridge Regression are also informative. Given that G1, G2, study time, and failures are all correlated in this dataset, one would expect Ridge's regularization to produce a more noticeable improvement. The fact that the difference is so small suggests that the multicollinearity is not strong enough to cause serious instability in the OLS solution. For institutions where simplicity and interpretability are priorities, standard Linear Regression is a perfectly acceptable choice with negligible accuracy cost compared to Ridge Regression.

Poor performance of the individual Decision Tree algorithm provides the key takeaway from this problem in terms of machine learning methodologies. R^2 score equals to 0.6817, which is more than 14 percentage points behind that of Random Forest. In turn, MSE of 5.190 is almost twice as large compared to the MSE of Random Forest – 2.838. This huge difference between the two metrics is explained by the fact that individual Decision Tree becomes so deeply trained that it memorizes all 316 training observations. Individual Decision Tree is able to recognize patterns of training observations but does not generalize well to the test sample of 79 students. The Random Forest algorithm addresses this problem through each individual tree committing the memory of distinct random subsets of data and subsequently averaging out these memories, thus creating a much more consistent model.

The feature importance findings carry the most direct practical implications. The domination of G2 and G1 in the predictor list makes one thing very clear: the single most valuable input for any student performance prediction model is simply reliable records of students' actual academic performance during the year. Schools do not need to collect complicated survey data to build a capable early warning system. If they have reliable grade data, they already have a very strong prediction foundation. The most important takeaway for school administration is to invest in a good mid-year tracking system and feed that data into the prediction model as early as possible.

Other indicators that were considered to be important include absences and prior failures, which are inputs that the schools have the capability of monitoring and addressing immediately. For instance, a high number of absences by a student is a behaviour indicator, indicating an early-warning sign and may necessitate a pastoral care discussion with the student. If a student has been unsuccessful in his courses before, then he or she qualifies for an early-warning indicator. In summary, an ideal early-warning system would include the model prediction as the first step towards an informative dialogue with the student.

However, one major drawback was identified during the analysis. The accuracy of the predictions for extremely low performing students, especially for those with a score of 0, is relatively lower compared to other scores. These are actually the students that require urgent attention. The explanation for this problem has to do with the rare occurrence of such samples in the dataset. Due to the limited number of observations of zero-grade students, it was impossible for the algorithm to learn their patterns.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study focused on evaluating the efficiency of four well-known machine learning regression models in predicting the final grades of students based on the UCI Student Performance Dataset. Four machine learning regression models – Linear Regression, Ridge Regression, Decision Tree Regressor, and Random Forest Regressor were analysed in terms of their accuracy in predicting the final grade of students using an 80:20 train-test split, data loading, data exploration, categorical encoding via LabelEncoder, normalization via StandardScaler, model training.

Results were clear and consistent. The best performing model was the Random Forest Regressor which had an R^2 score of 0.8288, MAE score of 1.147, and an MSE score of 2.838. It proved how powerful the use of ensemble modelling could be. The second-best model was the Ridge Regression model which had an R^2 score of 0.7625 followed by the Linear Regression model with an R^2 score of 0.7614. Lastly, the Decision Tree Regressor model came fourth with an R^2 score of 0.6817.

From the feature importance analysis, we can conclude that the top two factors are G2 and G1 which have the highest predictive power for G3 followed by absence and failure. Thus, we can conclude that intermediate academic performance is the only factor with the highest value for constructing an early warning system in schools.

Based on the results obtained from the research, we can say that the best technique for predicting the performance of students in educational institutions would be ensemble machine learning techniques, where Random Forest can be considered an optimal algorithm due to its reliability and interpretability.

B. Future Work

Various research directions can be derived from this study. Firstly, it will be interesting to explore more advanced models like XGBoost, LightGBM, and Gradient Boosting compared to the Random Forest model already built, in order to see if accuracy can be further improved. The Deep Learning techniques, for example multi-layer perceptrons, can also be considered but they generally need more data to perform well.

Additional improvements in the performance of the Random Forest algorithm can be made through systematic tuning of its hyperparameters through grid search, random search, or Bayesian optimization techniques other than the default ones utilized in this experiment. The use of k-fold cross-validation, as opposed to the one-time splitting of data into an 80:20 ratio for training and testing, respectively, can be beneficial.

It is worthwhile to spend more resources on a better statistical approach to evaluating the model. The single train-test split used here introduces some variance depending on which students end up in the test set. With k-fold cross validation, where the data is divided into k equally sized sets and the process repeats k times, the results would be much more stable.

To tackle the problem of prediction failure for zero grade students, further study may include investigation into the use of oversampling with SMOTE to artificially synthesize training samples of low-grade students who are in the minority, or cost-sensitive learning where more weight is attached to misclassification of critical groups. Lastly, moving from development of predictive models into implementation through actual application in practice, for instance by developing a dashboard that provides prediction and risk factors per student, would be highly beneficial.

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